



11.7 Fundamental Constituents of Matter and Their Interactions

Sri Lanka Atomic Energy Board

Particle Physics



- The study of the fundamental particles of the universe and interaction among them
- Building blocks of universe are called as fundamental building blocks
- A particle is considered to be elementary only if there is no evidence that it is made up of smaller constituents

Experimental approach probing the structure of matter

- In ancient time, the Greek thinker **Empedocles** first classified the fundamental elements as;
 - Fire
 - Air
 - Earth
 - Water



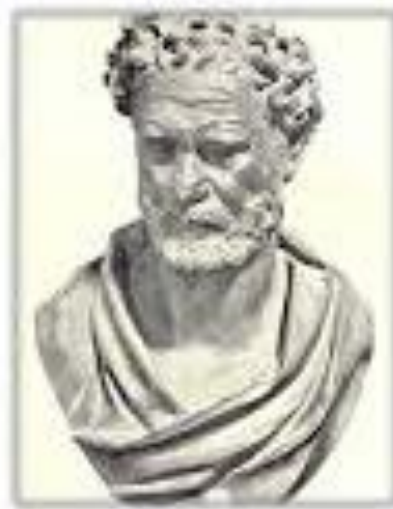
ATOM

- Ancient Greek philosophers of Leucippus and his pupil Democritus introduced the word “Atom” based on ideas were founded in philosophical and theological reasoning rather than evidence and experimentation.

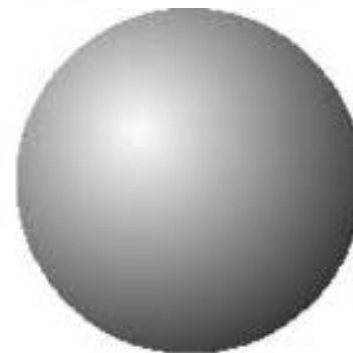
Leucippus



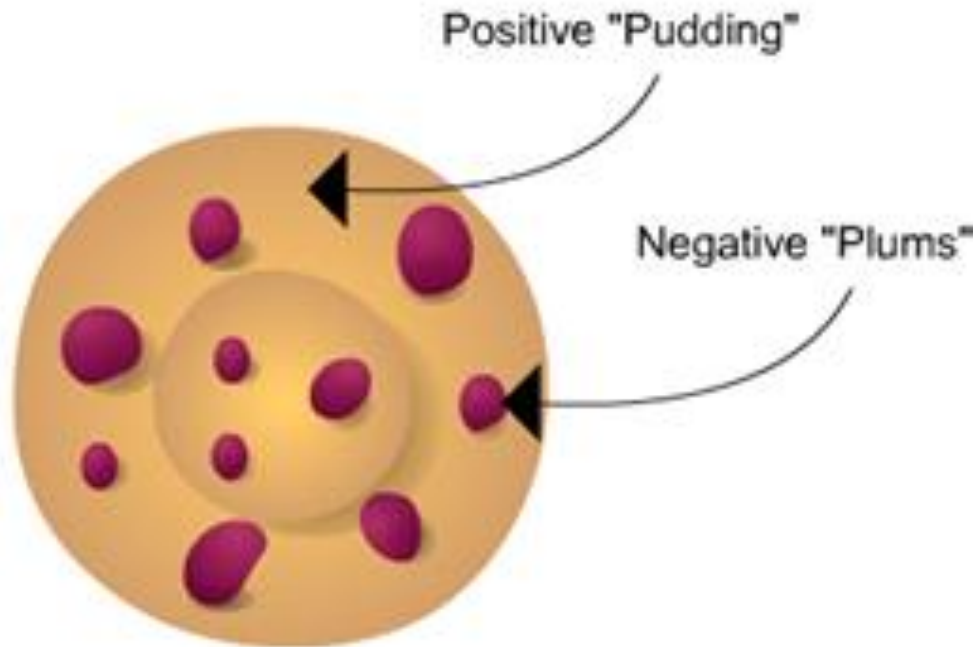
Democritus



THE ATOM



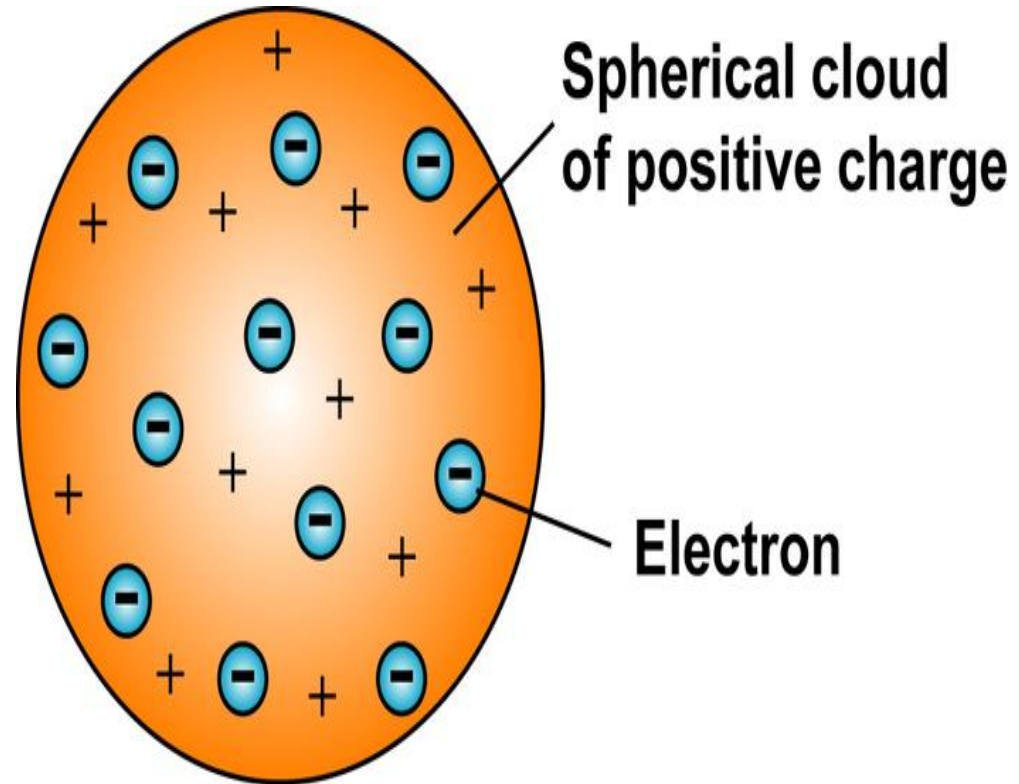
J. J. Thompson's "Plum Pudding" Model



- Discovery of the electron was done by Mr. J.J. Thompson during year 1897. Therefore, atom is not a fundamental matter any more
- He introduced a concept for Atom, which was called as "plum pudding" model

“Plum pudding” model

- Electrons are very tiny, electrically negative charge particles. These particles are embedded in the positively charged medium
- Positive charge of the medium balance the negative charge and Atom remains as neutral.

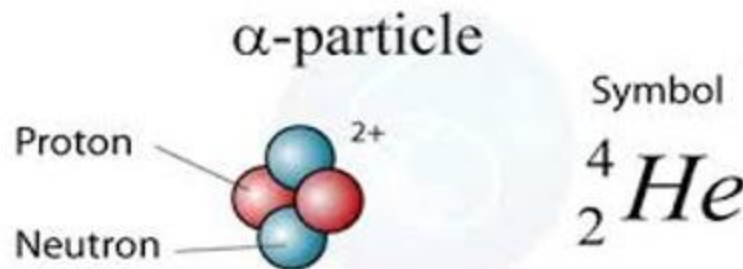


Rutherford's Gold Foil Experiment

- Continuous studies carried out from 1909 to 1911 and discovered that every atom contains a nucleus.

Gold Foil Experiment

- They pointed a beam of alpha particles at a thin foil of gold and measured the scattering pattern by using a fluorescent screen

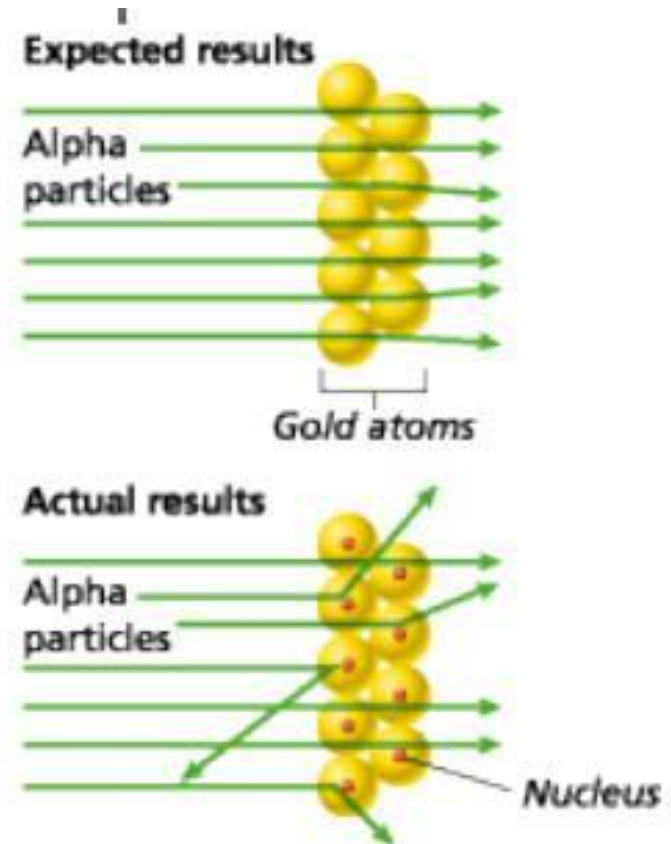
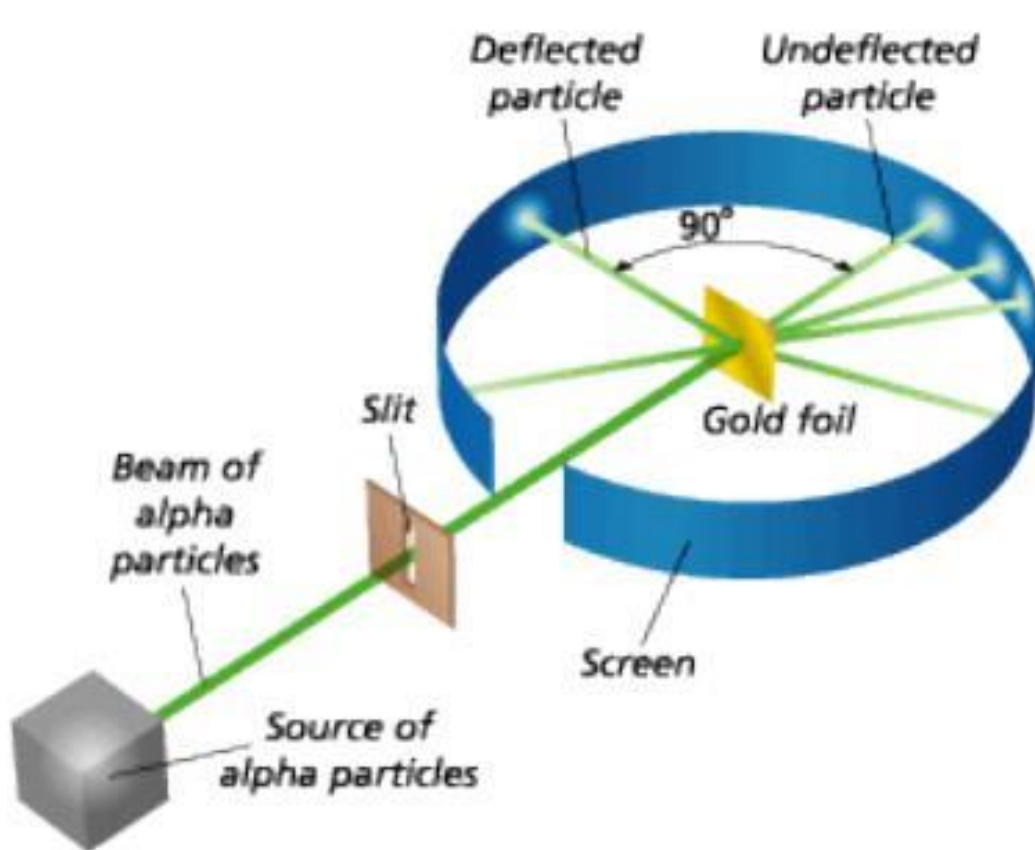


Alpha particle is nucleus of helium

Rutherford's Gold Foil Experiment

- Objective of the experiment was to study the distribution of electrons in the atoms' Plum Pudding Model.
- Rutherford expected very little deflection of the path of the alpha particles due to electrons of the model.
- Noticed only a tiny fraction of the alpha particles were deflected by more than 90° . Most flew straight through the foil.

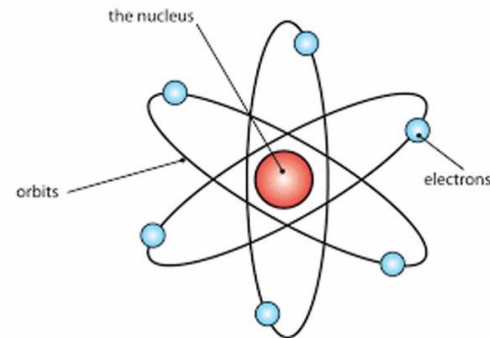
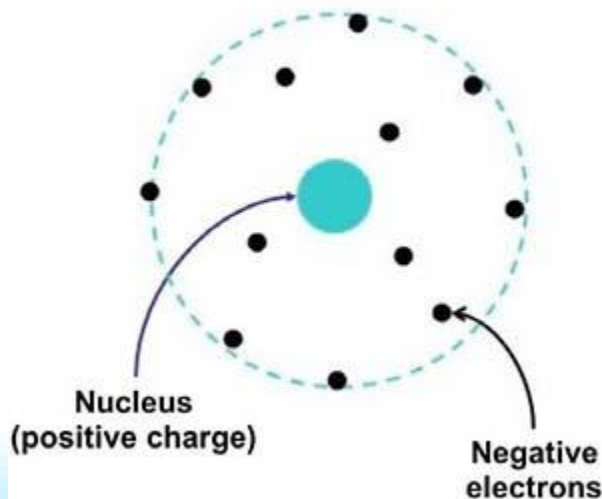
Rutherford's Gold Foil Experiment



Rutherford's Gold Foil Experiment

- Rutherford thus rejected Thomson's model of the atom.
- Proposed a model where the atom consisted of mostly empty space, with all its positive charge concentrated in its center in a very tiny volume, surrounded by a cloud of electrons.

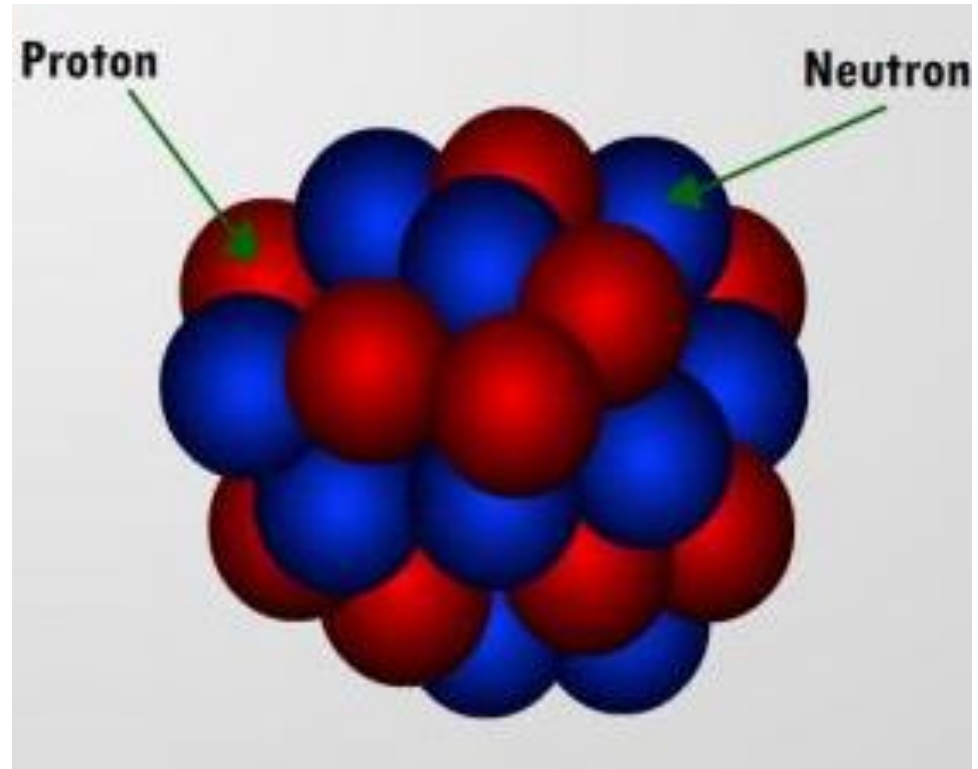
RUTHERFORD'S ATOMIC MODEL



Rutherford's Model Of An Atom

Is nucleus fundamental particles?

- Nucleus considered fundamental particle as it is very tiny and dense unit.
- In 1932, James Chadwick discovered neutron.
- Therefore, nucleus consist of neutrons and protons



Requirement of high energy particles to study the structures of matter

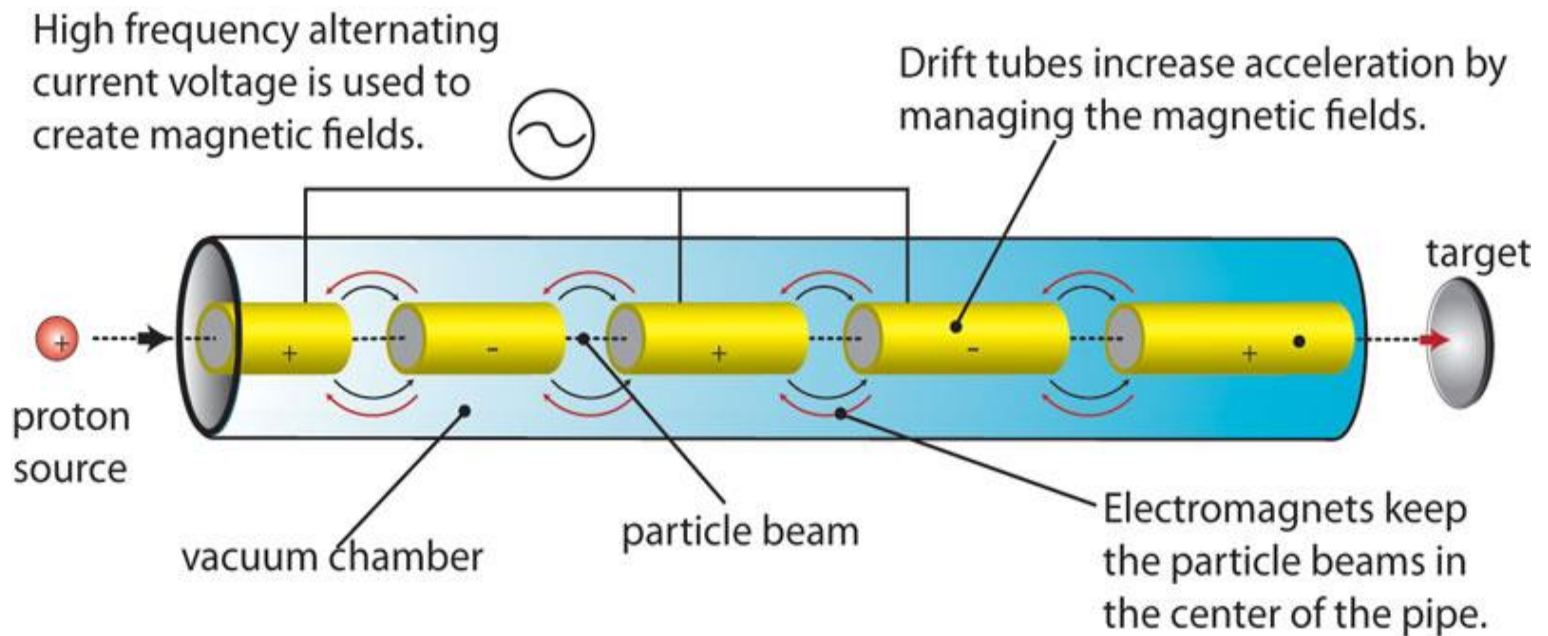
- Very tiny, high energy particles are required to carry out studies on the building blocks of protons and neutrons
- According to the De Broglie hypothesis, resolution power increase with decrease of wavelength

$$\lambda \propto 1/mv$$

- In order to increase resolution, mass and velocity should increase
- For a particle, mass is constant, therefore velocity should increase in order to increase resolution.
 - $E = (1/2) mv^2$
 - $v = \sqrt{2E/m}$
- Velocity increase with the increment of the energy.
- Therefore, high energy particles are required to study very tiny particles.
 - Energy (E) = Force on charged particles which are available in the electric field x Distance (d)
- Production of high energy particles can be done by accelerating charged particle in a electric field, into nearly speed of light.

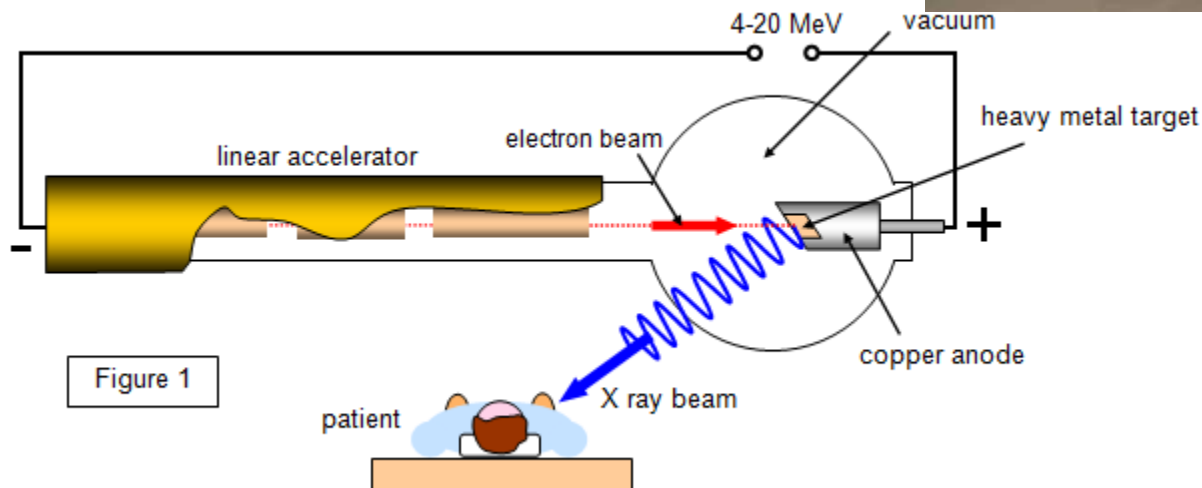
Particle Accelerator

- Charged particles are accelerated into high velocity. Here alternating current voltage is used to create magnetic field.
- These accelerated particles collide with another particles in order to conduct researches.



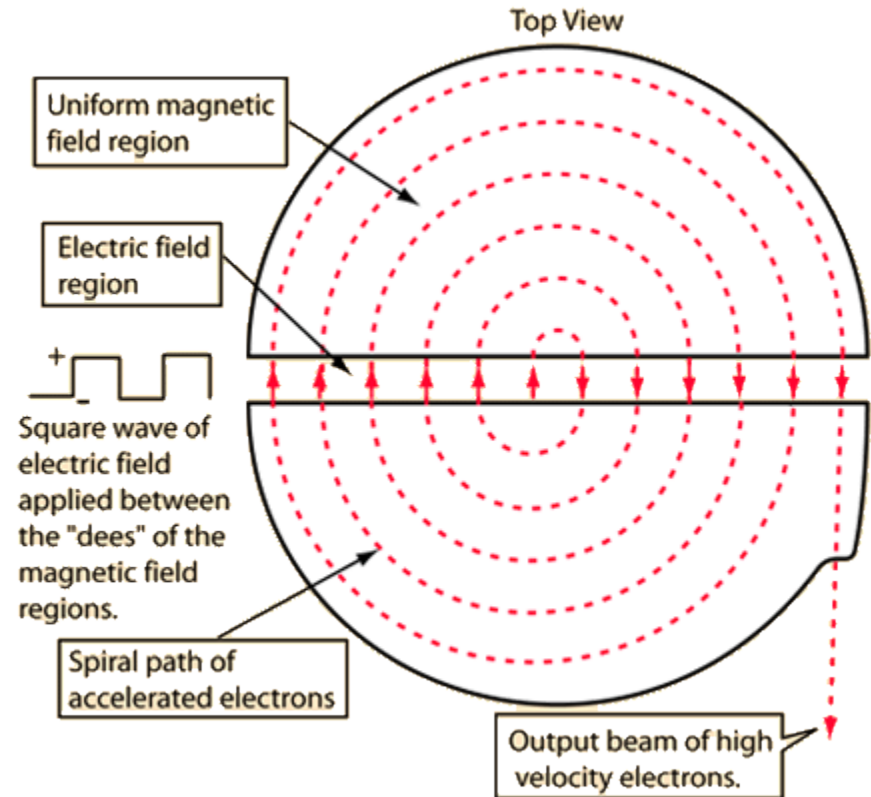
Linear Accelerator (Linac)

- Use to treat cancers found inside of the body. High energy beam or electric beam releases into the place where cancer cells present.



Cyclotron

- Cyclotron is consist of D shaped two electrodes, and alternating current supply.
- Charged particles travel through semi circular magnetic field. When these charge particles come into the electric field (in the gap between two D shaped electrodes) velocity of the particles increase.
- Cyclotrons use to produce high energy charged particles, radiopharmaceuticals, (PET scanner, Technetium-99m) and use for cancer treatments (photon therapy)



Particle accelerators are used to produced high energy particles

- Giant accelerators are used to conduct particle physics experiments in the world,.
- Hadron Collider(LHC) which is located in Switzerland is the most powerful accelerator.
- Produce high energy particles which use to carry out studies on fundamental particle physics.

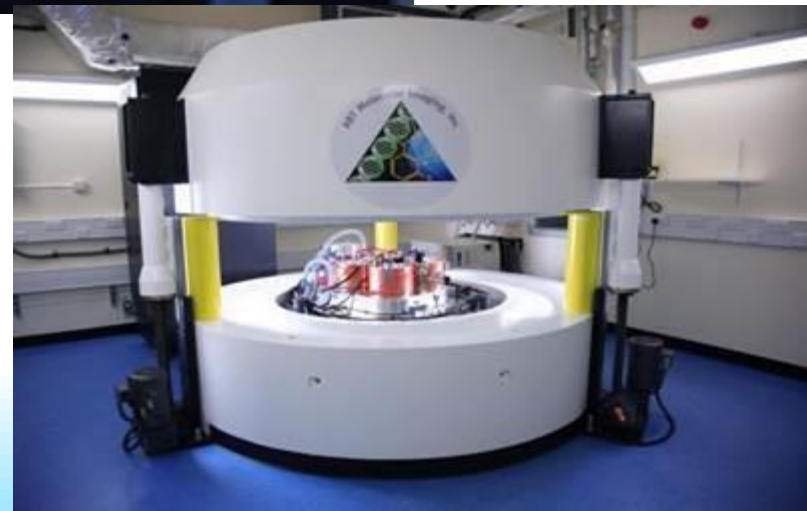
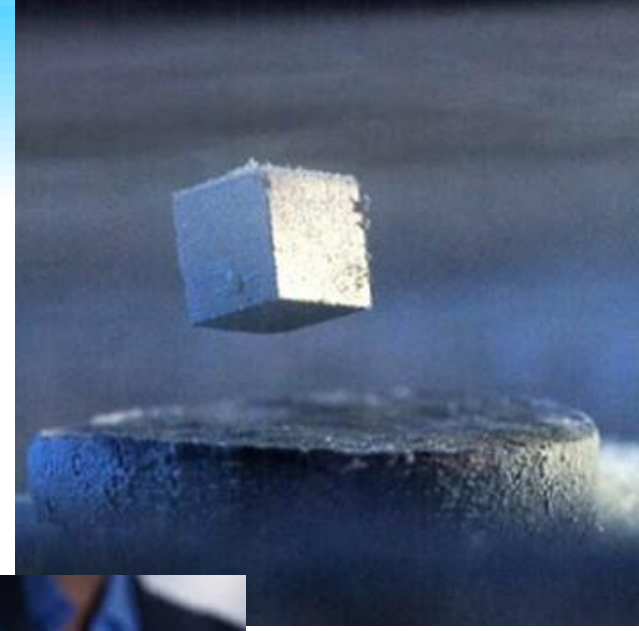


Aerial View of the Large Hadron Collider in Geneva



Applications of Particle Accelerators

- Produce super conductors
- Touch screen production
- Fields of industry and human health

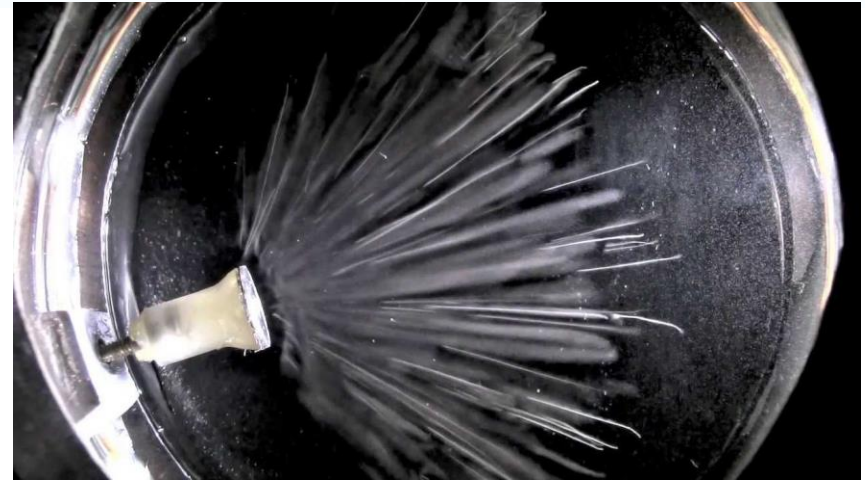


Particle detector

- **Particle detector**, also known as a **radiation detector**, is a device used to detect, track, and/or identify ionizing **particles**, such as those produced by nuclear decay, cosmic radiation, or reactions in a **particle** accelerator.
- Detectors can measure the particle energy and other attributes such as momentum, spin, charge, particle type

Types of Detector

- Historical examples for detectors-bubble chamber, cloud chamber
- Detectors for radiation protection - gaseous ionization, Scintillation and Semiconductor detectors.
- Detectors use for particle and radiation physics experiments. Ex: Solid-state detectors, Calorimeters, Neutron detectors



Gas-filled Geiger-Muller
Pancake Detector

NaI Solid Scintillation Detector

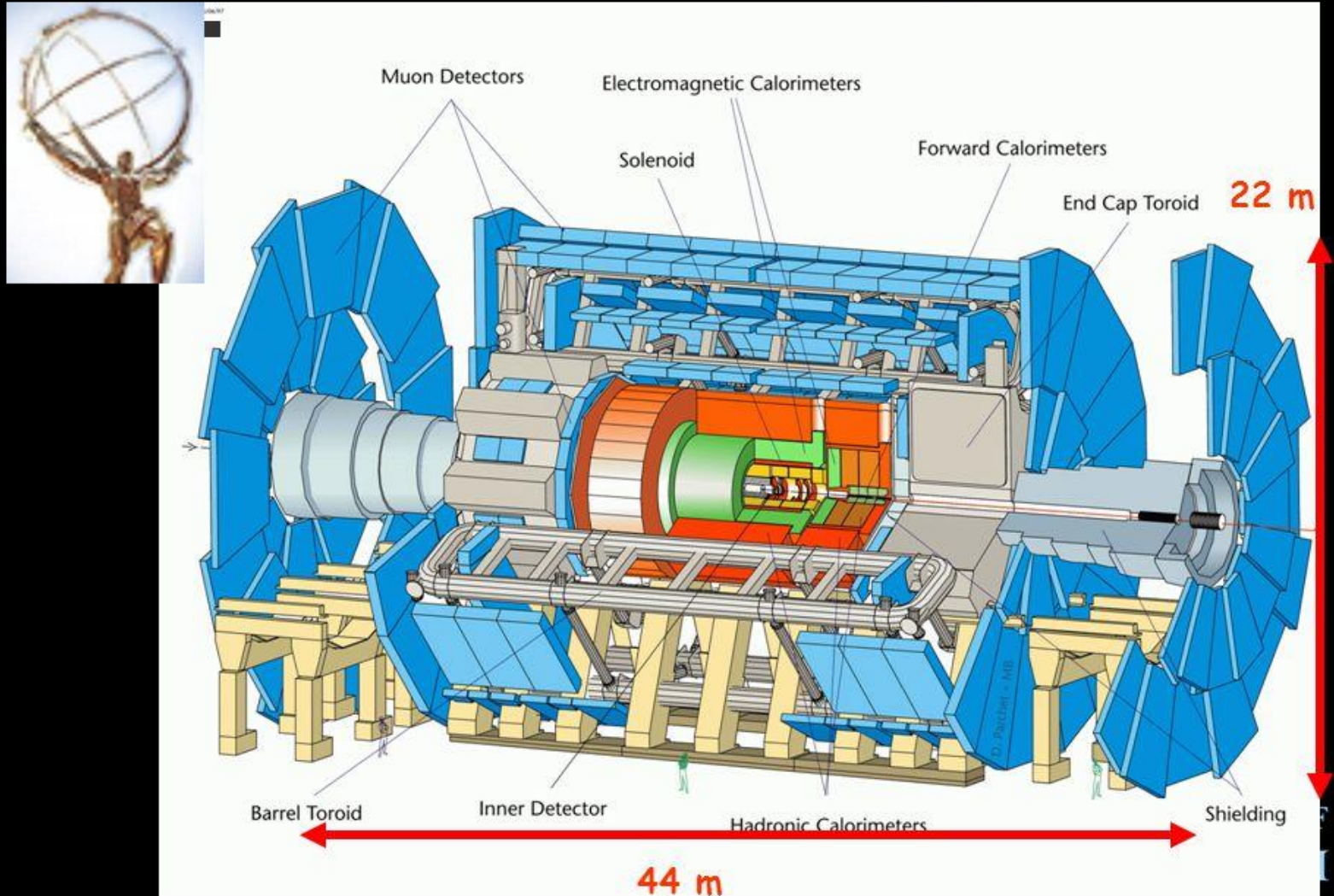


Detectors for Accelerators

- Detectors designed for modern accelerators are huge, both in size and in cost.
- The term *counter* is often used instead of *detector* when the detector counts the particles but does not resolve its energy or ionization. Particle detectors can also usually track ionizing radiation (high energy photons or even visible light). .
- Detectors of CERN – Large Hadron Collider (LHC)
 - CMS - Compact Muon Solenoid
 - ATLAS - A Toroidal LHC ApparatuS
 - ALICE - A Large Ion Collider Experiment
 - LHCb - Large Hadron Collider beauty

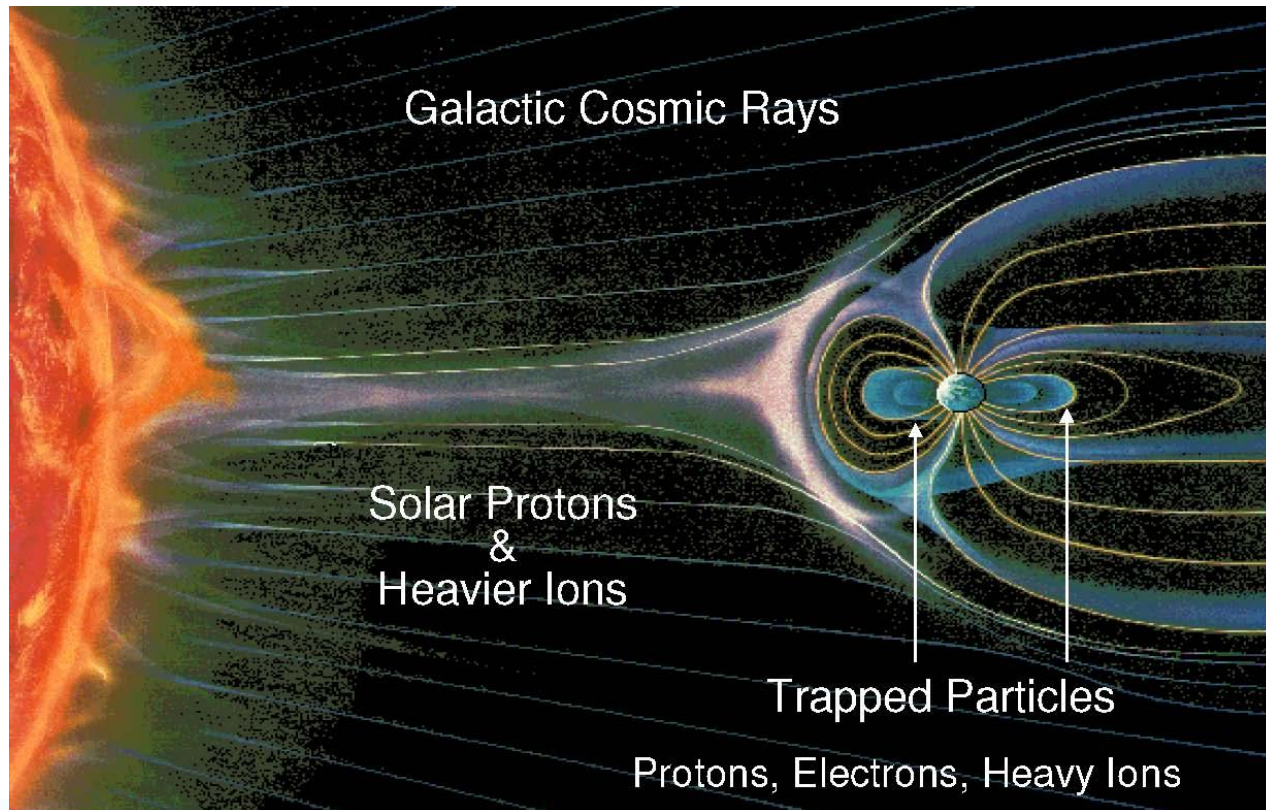
ATLAS - A Toroidal LHC ApparatuS

ATLAS: A Toroidal LHC ApparatuS

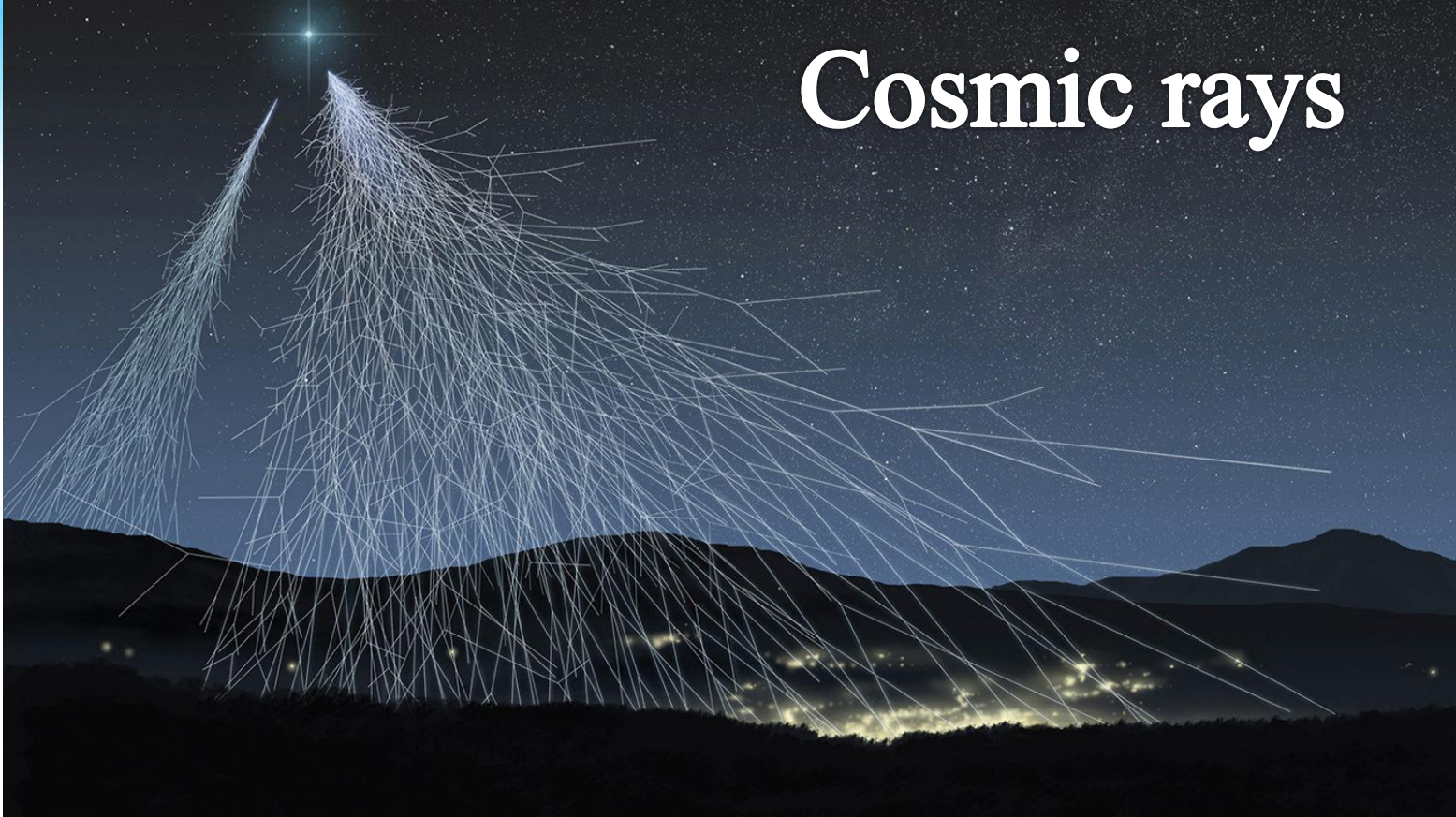


Cosmic rays

- High energy particles use to study structure of the matter
- **Cosmic rays** are high-energy radiation, mainly originating outside the Solar System and even from distant galaxies.

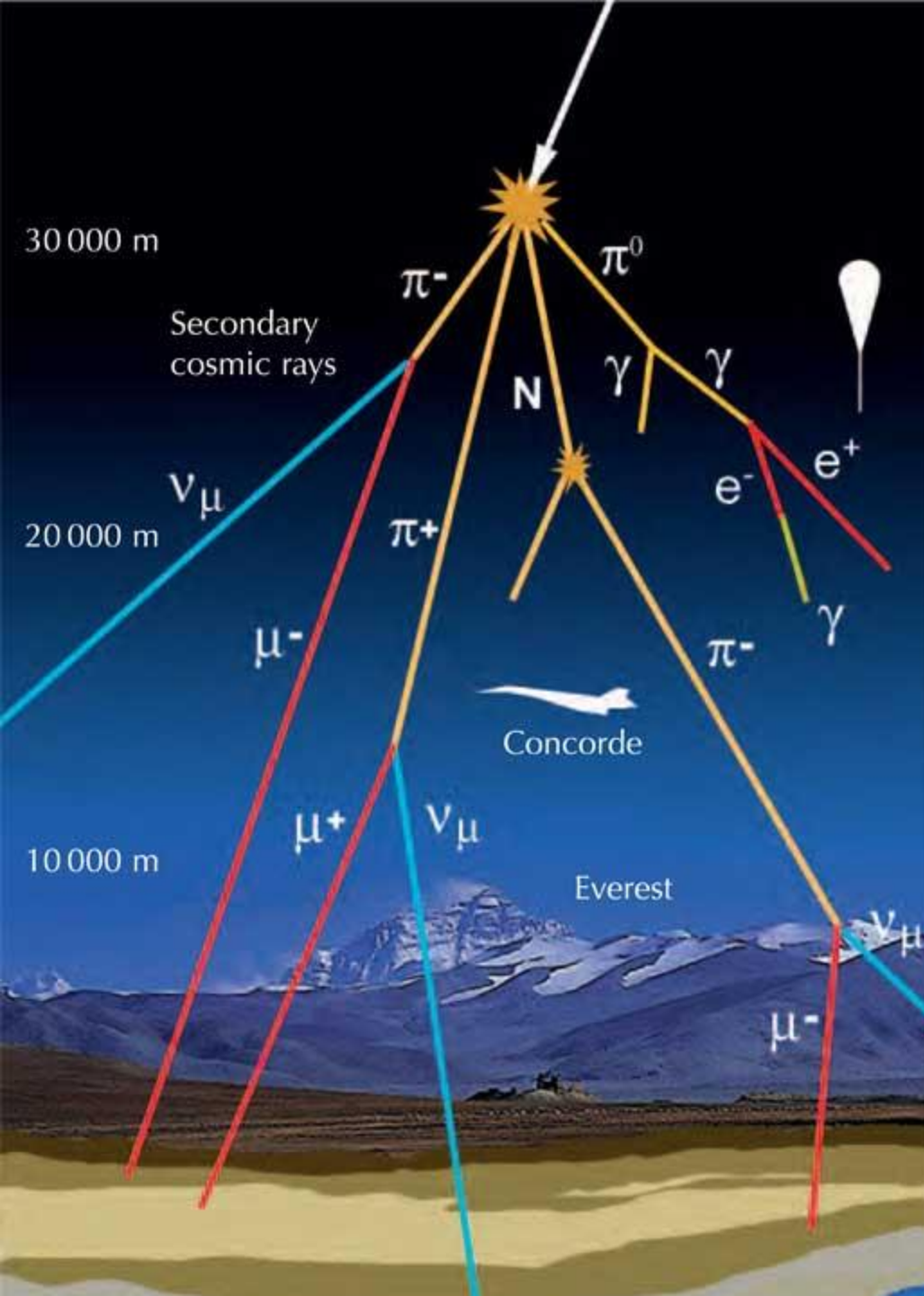


Cosmic rays



- Composition of the cosmic rays; Protons/ Hydrogen nuclei 89%, Helium nuclei 10% and heavy metal nuclei 1%.
- Speed of cosmic rays near to the speed of light.
- When cosmic rays enter into earths' atmosphere, they collides with atomic nucleus.

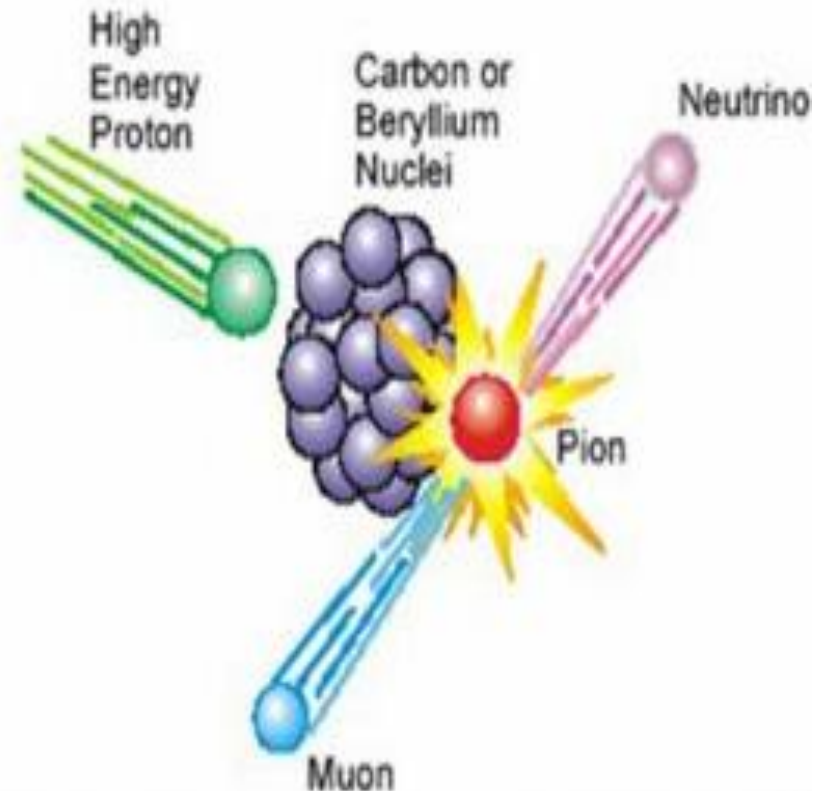
Cosmic rays



- Scientists who worked in Bristol university in 1947 have found PIONS.
- When cosmic rays entering into the atmosphere, they collide and produce high energy tiny particles named PIONS
- PIONS - $\pi^+ \pi^0 \pi^-$
- PIONS decay instantly.

Cosmic rays

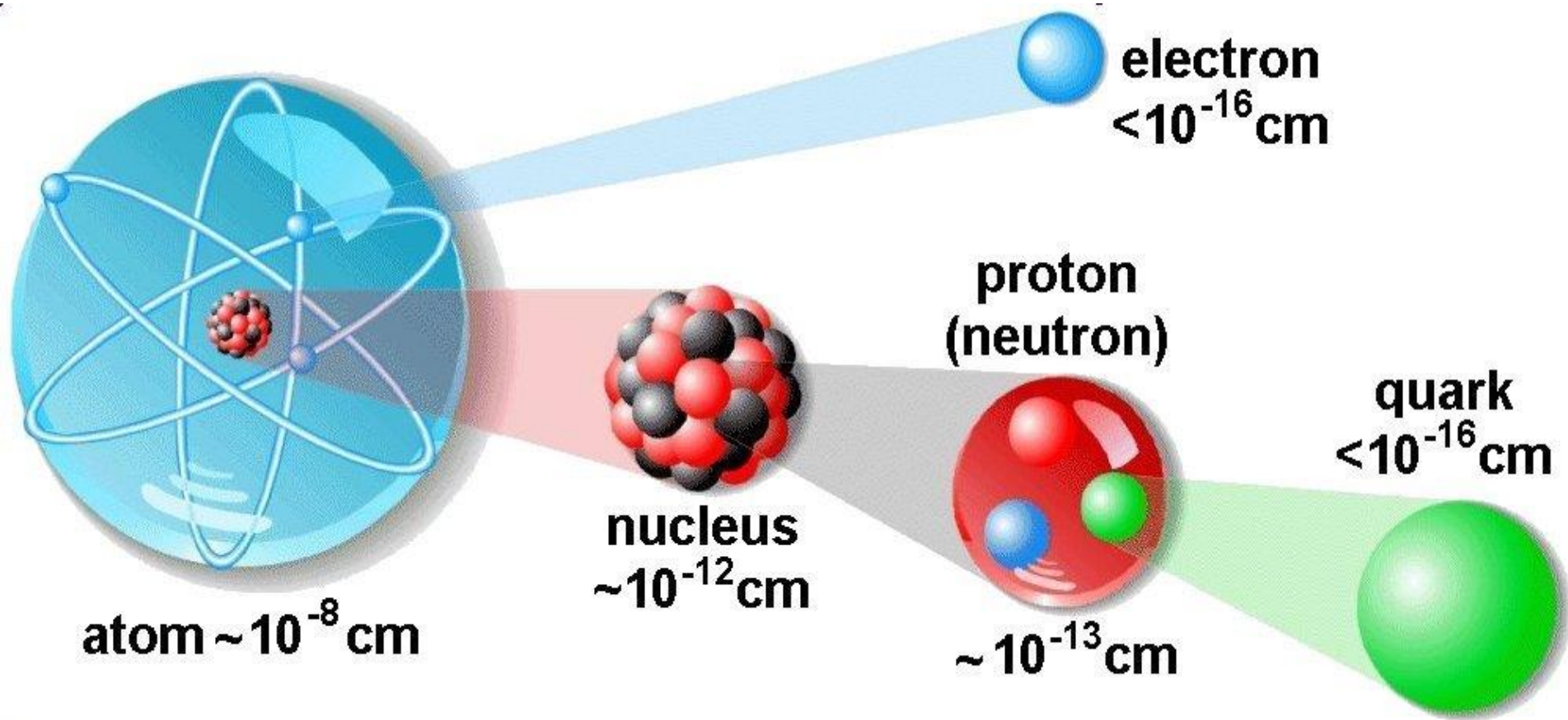
- Pion Decay
- Positive Pion (π^+) – Muon (μ^-)
- Negative Pion (π^-) – Muon Neutrino (ν_μ)
- Neutral Pion (π^0) – Gamma rays
- Muon doesn't collide with matter of the atmosphere, they travel through the atmosphere in high speed and come to earth, sometimes penetrate in to the earth surface.
- Muon can be found in the deepest ocean, underground and main contributor to the background radiation



Artificial Pion Production

- 1948 Eugene Gardner and his team artificially produced Pions in the university of California using particle accelerator.
- High energy alpha particle which accelerated into high velocity, collided with carbon atom and then produce Pion

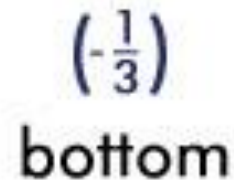
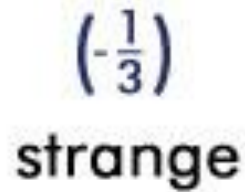
Fundamental Particles



Fundamental Particles

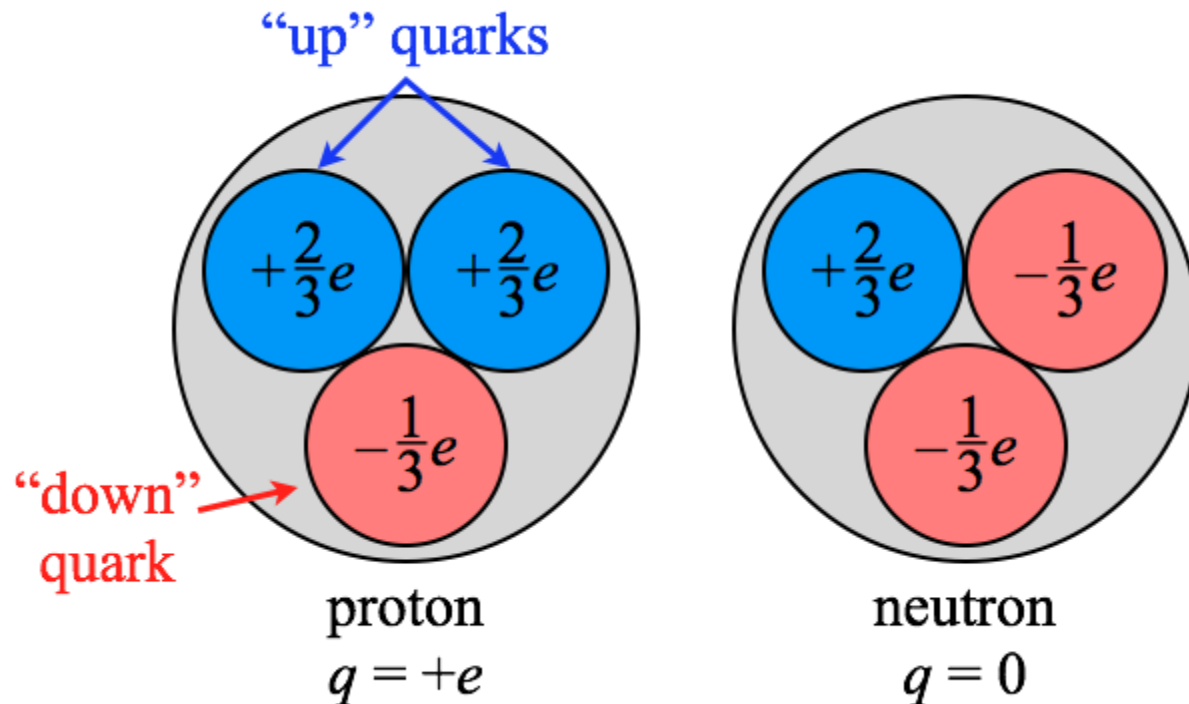
- Building blocks of Protons and Neutrons are quarks and these quarks can not divide further.
- Quarks and electrons are known as fundamental particles
- Quark model was developed by Murray Gell-Mann and George Zweig in 1964
- There are six types of quarks. Accelerators have detected all the six types and final one detected during year 1995 at the fermi LAB.

Quarks



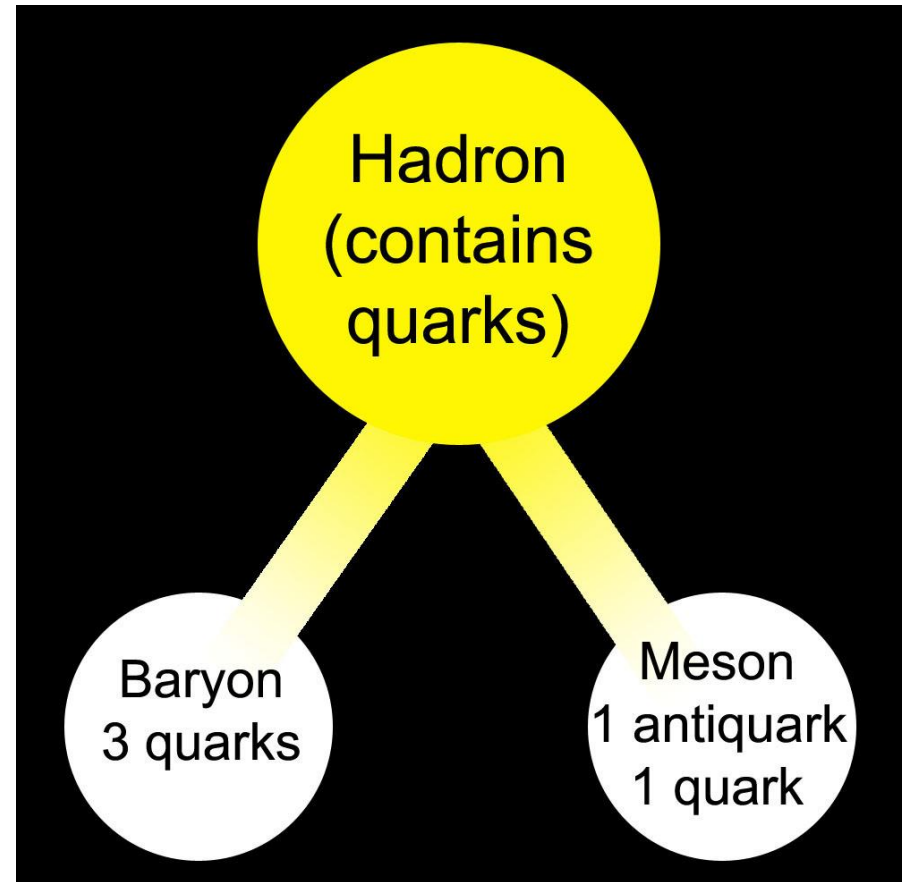
Proton and Neutron

- Both proton and neutron consist with three quarks.



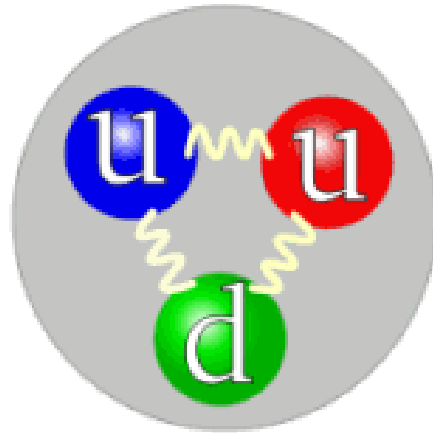
Hadrons, Baryons, Mesons

- Like social elephants, quarks only exist in groups with other quarks and are never found alone. Composite particles made of quarks are called Hadrons
- Although individual quarks have fractional electrical charges, they combine such that hadrons have a net integer electric charge..
- There are two classes of hadrons

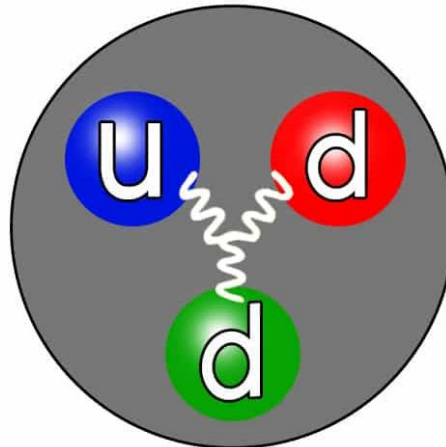


Baryons

- They are any hadron which is made of three quarks
- proton- 2 up quarks + 1 down quark (uud)

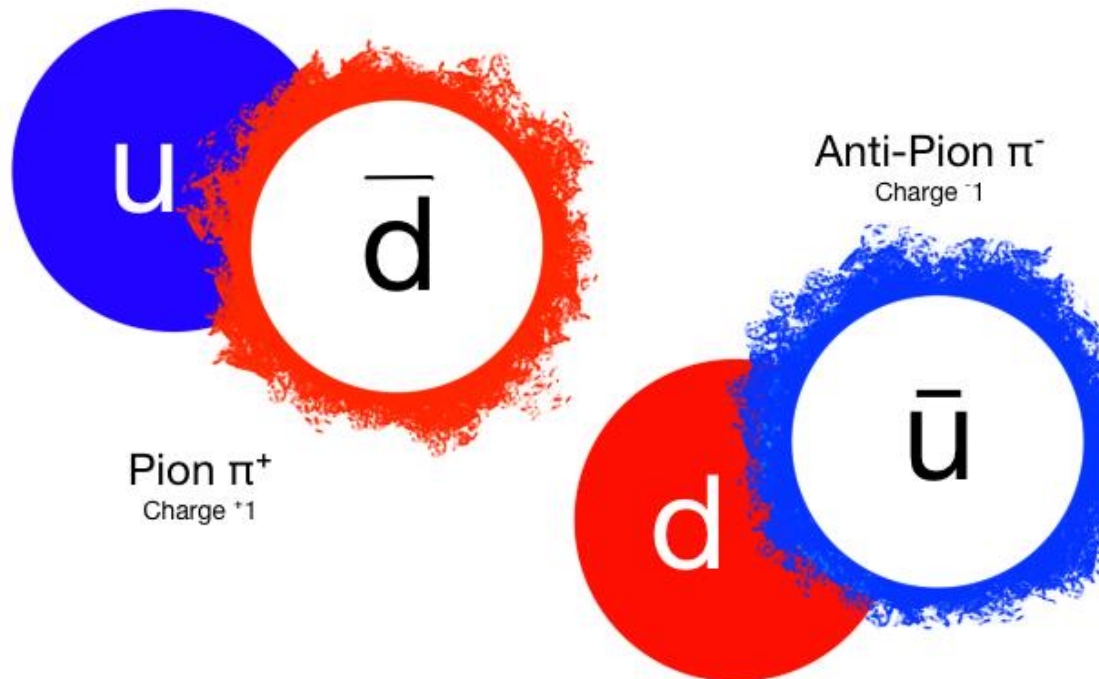


- neutron- 1 up quarks + 2 down quark (udd)



Mesons

- Mesons contain one quark (q) and one antiquark ($\bar{q}$)
- Ex: pion - made of an up quark and a down antiquark
- antipion - made of a down quark and an up antiquark.



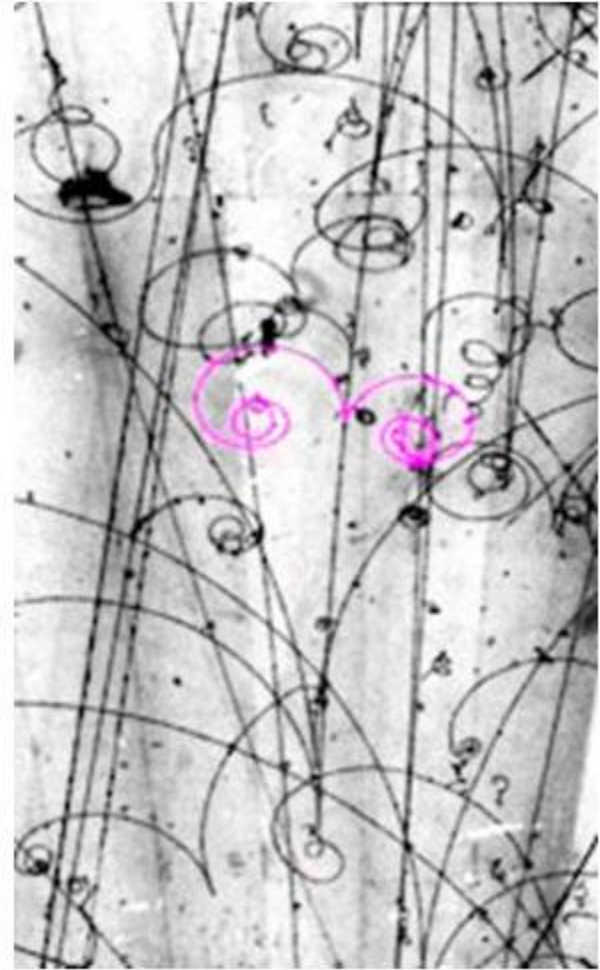
Matter and Antimatter

- For every type of matter particle we've found, there also exists a corresponding **antimatter** particle or **antiparticle**.
- Antiparticles look and behave just like their corresponding matter particles, except they have opposite charges
- A proton is electrically positive whereas an antiproton is electrically negative



Matter and Antimatter

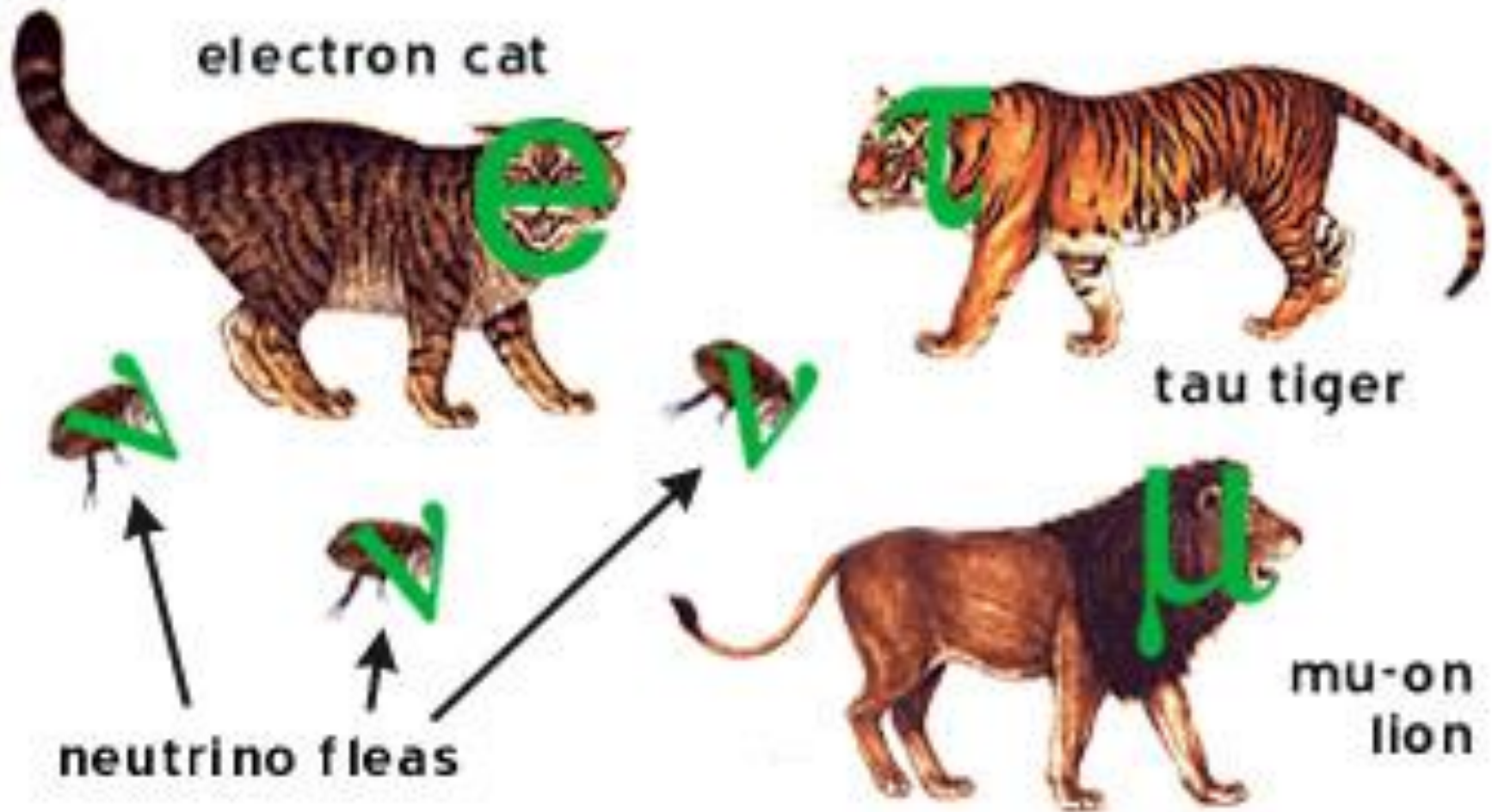
- Gravity doesn't affect for matter and antimatter as mass doesn't change
- The **magnetic field** in this chamber makes negative particles curl left and positive particles curl right.



Leptons

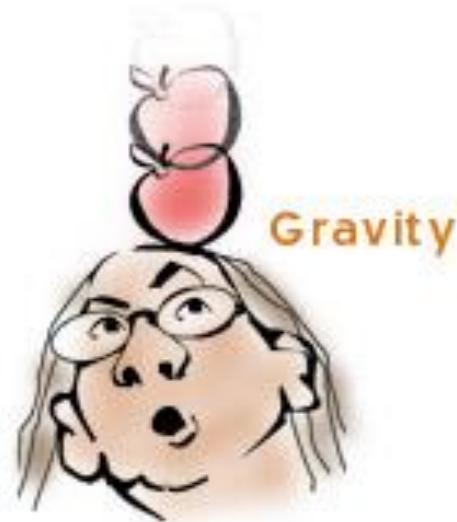
- The other type of matter particles are the **leptons**. They appear to be point-like particles without internal structure.
- There are six leptons, three of which have electrical charge and three of which do not
 - **Electron**(e^-)
 - **Muon** (μ^-)
 - **Tau**(τ^-)
 - **Electron neutrino** (ν_e)
 - **Muon neutrinos** (ν_μ)
 - **Tau neutrinos** (ν_τ)
- **Muon** and **tau**, which are charged like electrons but have a lot more mass⁴⁴
- The other leptons are the three types of **neutrinos**. They have no electrical charge, very little mass, and they are very hard to find
- Leptons are solitary particles. (Independent Cats associated with neutrino fleas)

Leptons



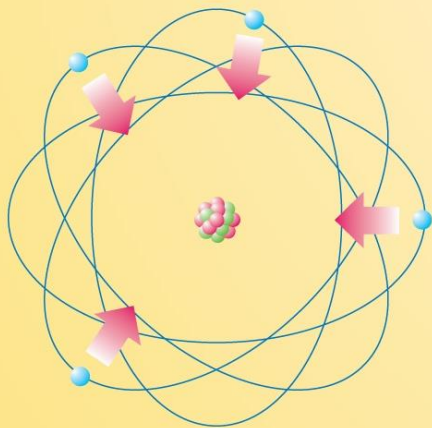
Fundamental Force

- Gravitational force
- Electromagnetic force
- Strong force
- Weak force
- Fundamental forces are important to bind fundamental particles together



The Four Fundamental Forces of Nature

**Electro-
magnetism**



**Weak
Interaction**

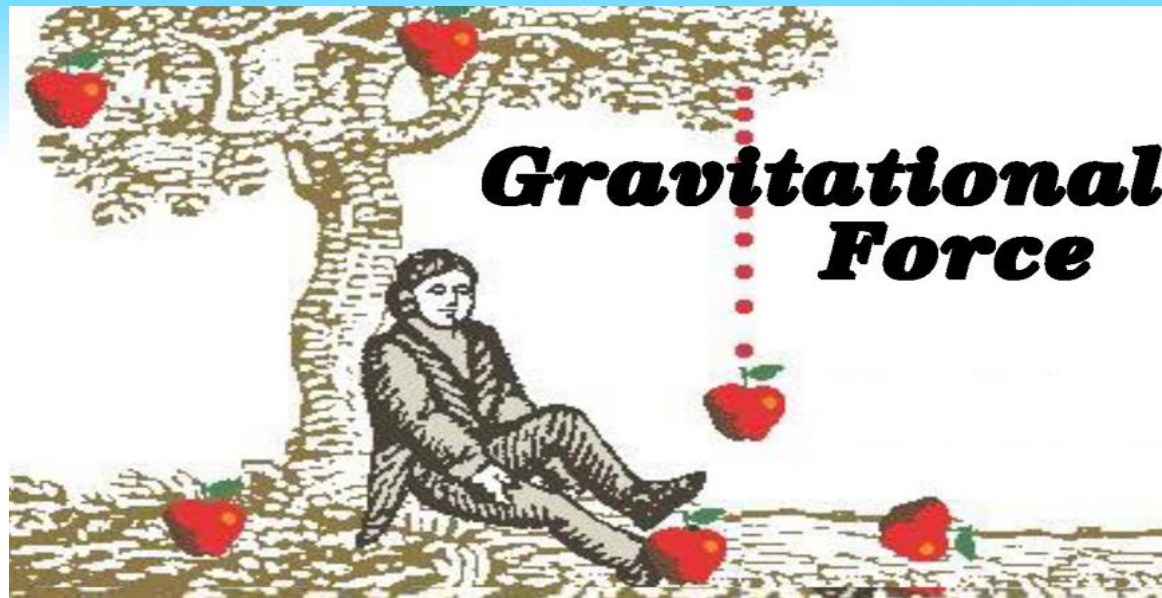


**Strong
Interaction**



Gravitation





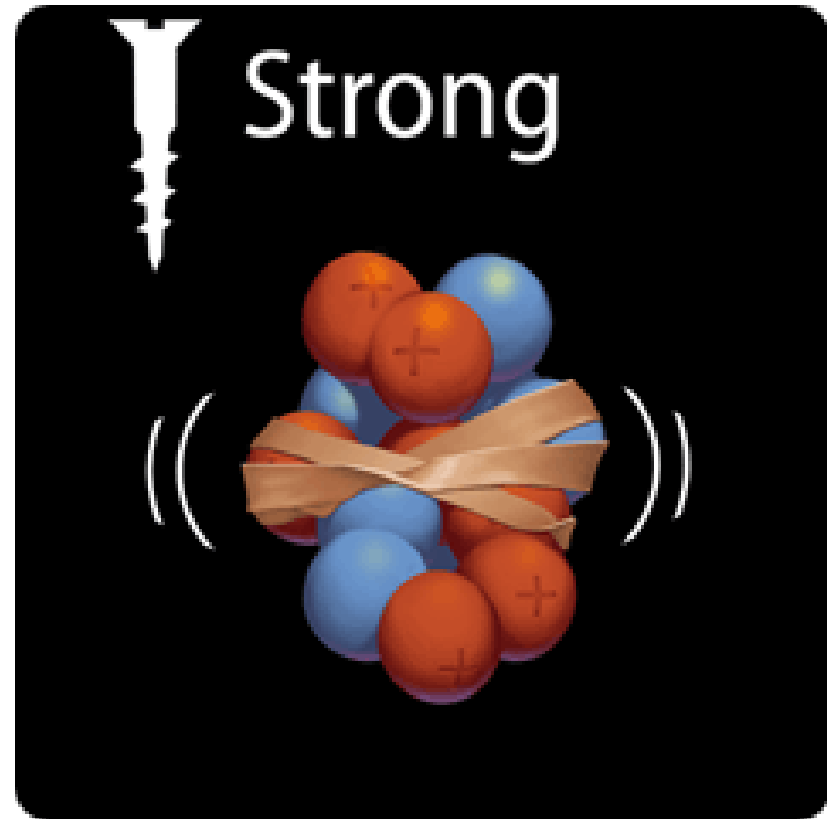
- Natural Phenomena
- All the object which having mass are subjected to gravitational force
- Weakest force among four fundamental forces
- Gravitation matters on the large-scale structure of the universe and gravitational force for sub atomic particles are can not be considered.

Electromagnetic force

- Can be seen among charge particles as well as electromagnetic field
- The electromagnetic force plays a major role in determining the internal properties of most objects encountered in daily life
- The electromagnetic force governs all chemical processes, which arise from interactions between the electrons of neighboring atoms.
- Electromagnetic force is what allows atoms to bond and form molecules, allowing the world to stay together and create the matter you interact with all of the time
- Electromagnetic force is stronger than gravitational force
- The carrier particle of the electromagnetic force is the **photon** ().

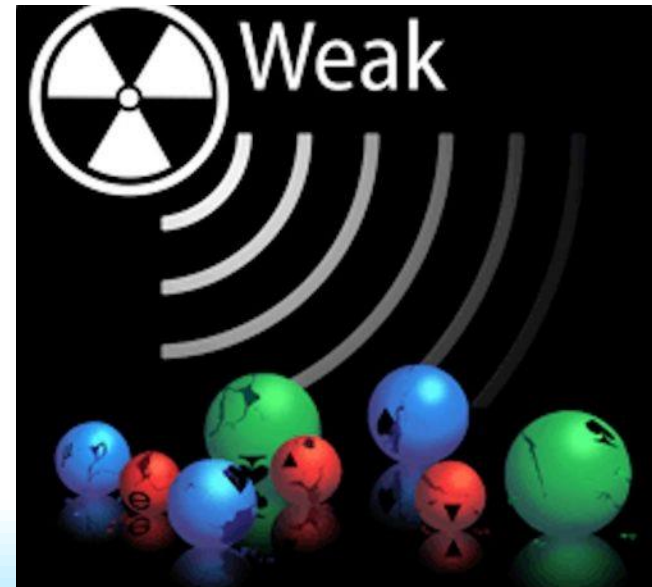
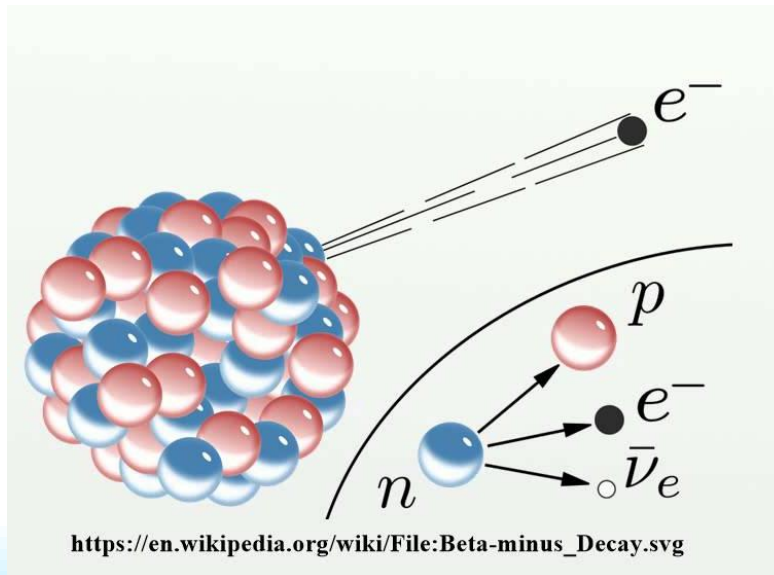
Strong Force

- Can be seen in between sub atomic particles and this force responsible for nuclear interactions
- The strong force holds quarks together to form hadrons and force carrier particles are gluon



Weak Force

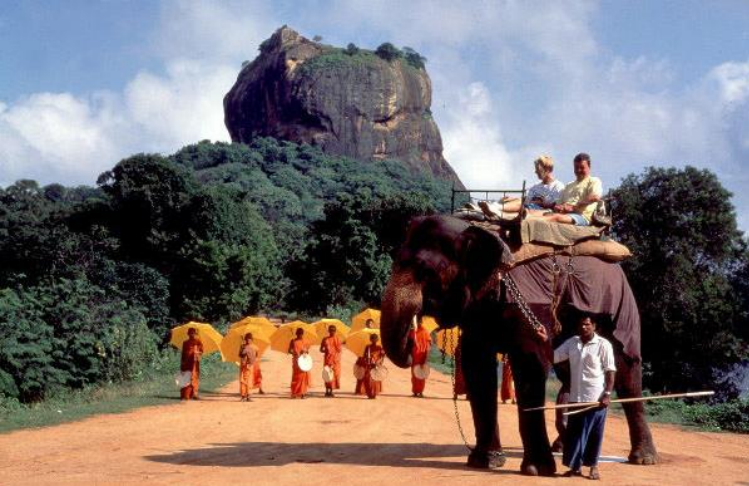
- Can be seen among sub atomic particles and responsible for radioactive decay .
- Force carrier particles are W and Z Bosons



Standard Model

Elementary Particles

Quarks	u up	c charm	t top	γ photon
	d down	s strange	b bottom	
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	
Leptons	e electron	μ muon	τ tau	Z Z boson
				W W boson
I II III				Force Carriers
Three Families of Matter				



Thank You

